



MBA PROGRAM

BUSA693: OPERATIONS MANAGEMENT

PROJECT1

**Comparative Analysis of Time Series Forecasting Methods
for Monthly Retail Sales**

Prepared by:

Mohammad Abu Shams	1255049
Issa M. Dahu	1255228
Nancy Morrar	1255087
Ahmad Aburahma	1255082
Khader Fawadleh	1245174

Instructor: Dr. Samir Baidoun

Section: 1

Date: 21-9-2026

Table of Contents

List of Figures.....	ii
List of Tables	iii
Introduction.....	1
Data Description.....	1
Forecasting Methods.....	4
1. Naive Method.....	4
2. Moving Average.....	5
3. Weighted Moving Average	7
4. Simple Exponential Smoothing.....	9
5. Trend-Adjusted Exponential Smoothing (FIT).....	11
6. Trend Projection	14
Seasonality Analysis.....	15
Combined Trend and Seasonal Forecast	17
Final Comparison of Forecasting Methods	19
Conclusion and Recommendation	22
References	23

List of Figures

Figure 1: Monthly Retail Sales from 2022 to 2025	3
Figure 2: Actual Sales and Naive Forecast	4
Figure 3: Actual Sales vs 3-Month Moving Average	5
Figure 4: Actual Sales vs 6-Month Moving Average	6
Figure 5: Actual Sales vs 12-Month Moving Average	6
Figure 6: Actual Sales vs Weighted Moving Average (0.50, 0.30, 0.20)	7
Figure 7: Actual Sales vs Weighted Moving Average (0.60, 0.30, 0.10)	8
Figure 8: Actual Sales vs Weighted Moving Average (0.70, 0.20, 0.10)	8
Figure 9: Actual Sales vs Exponential Smoothing ($\alpha = 0.10$)	9
Figure 10: Actual Sales vs Exponential Smoothing ($\alpha = 0.30$)	10
Figure 11: Actual Sales vs Exponential Smoothing ($\alpha = 0.50$)	10
Figure 12: Actual Sales vs FIT ($\alpha = 0.10$, $\beta = 0.10$)	11
Figure 13: Actual Sales vs FIT ($\alpha = 0.30$, $\beta = 0.20$)	12
Figure 14: Actual Sales vs FIT ($\alpha = 0.50$, $\beta = 0.30$)	12
Figure 15: Actual Sales vs FIT ($\alpha = 0.70$, $\beta = 0.50$)	13
Figure 16: Actual Sales vs Trend Projection	14
Figure 17: Monthly Seasonal Indices	15
Figure 18: Combined Trend and Seasonal Forecast vs Actual Sales.....	18
Figure 19: MAD Comparison Across Forecasting Methods	20
Figure 20: MSE Comparison Across Forecasting Methods	20
Figure 21: MAPE Comparison Across Forecasting Methods.....	21

List of Tables

Table 1: Monthly Retail Sales Data (2022–2025)	1
Table 2: Naive Method Forecast Accuracy	4
Table 3: Moving Average Results	5
Table 4: Weighted Moving Average Results.....	7
Table 5: Simple Exponential Smoothing Results	9
Table 6: Trend-Adjusted Exponential Smoothing Results	11
Table 7: Trend Projection Results.....	14
Table 8: Monthly Seasonal Indices.....	15
Table 9: Seasonal Ratios and Monthly Seasonal Indices.....	16
Table 10: Combined Trend and Seasonal Forecast – 2025.....	17
Table 11: Combined Trend and Seasonal Forecast Accuracy	17
Table 12: Final Comparison of All Tested Forecasting Methods.....	19
Table 13: Best Setting from Each Forecasting Method	19

Introduction

Forecasting is important in operations management because it helps organizations plan for future demand and make better decisions. Good forecasting can support inventory planning, purchasing, capacity planning, and the use of resources. Time series forecasting is useful when historical data are available over regular periods of time.

This project uses monthly retail sales data from January 2022 to December 2025. Several forecasting methods are used, such as the Naive Method, Moving Average, Weighted Moving Average, Simple Exponential Smoothing, Trend-Adjusted Exponential Smoothing, and Trend Projection. Different parameter values are tested for the main forecasting methods. The final comparison includes Moving Average, Weighted Moving Average, Simple Exponential Smoothing, and Trend-Adjusted Exponential Smoothing (FIT).

The accuracy of the forecasting methods is measured using Mean Absolute Deviation (MAD), Mean Squared Error (MSE), and Mean Absolute Percentage Error (MAPE). The main measure used to choose the best method is MAD, where the method with the lowest MAD is considered the best. Seasonality is also analyzed to understand the monthly patterns in the sales data, and a combined trend and seasonal forecast is developed.

The main goal of this project is to compare the forecasting methods, identify the most accurate method for the data, and provide a recommendation based on the results.

Data Description

The data used in this project are monthly retail sales data from January 2022 to December 2025. The dataset has 48 monthly observations. The data were taken from the Federal Reserve Economic Data (FRED) database, and the original source is the U.S. Census Bureau. The sales are measured in millions of dollars and are not seasonally adjusted.

Table 1: Monthly Retail Sales Data (2022–2025)

T	Month	Sales (Millions of Dollars)
1	2022-01-01	505782
2	2022-02-01	496221
3	2022-03-01	584512
4	2022-04-01	582200
5	2022-05-01	601192
6	2022-06-01	594419
7	2022-07-01	584625
8	2022-08-01	597265
9	2022-09-01	562463
10	2022-10-01	581106

11	2022-11-01	588876
12	2022-12-01	637438
13	2023-01-01	535580
14	2023-02-01	516582
15	2023-03-01	589560
16	2023-04-01	574331
17	2023-05-01	615939
18	2023-06-01	597476
19	2023-07-01	591150
20	2023-08-01	613511
21	2023-09-01	579306
22	2023-10-01	594266
23	2023-11-01	608426
24	2023-12-01	657243
25	2024-01-01	539834
26	2024-02-01	544169
27	2024-03-01	597695
28	2024-04-01	593774
29	2024-05-01	631832
30	2024-06-01	592999
31	2024-07-01	618432
32	2024-08-01	622421
33	2024-09-01	579083
34	2024-10-01	622738
35	2024-11-01	630399
36	2024-12-01	691312
37	2025-01-01	566662
38	2025-02-01	543970
39	2025-03-01	624954
40	2025-04-01	623756
41	2025-05-01	647306
42	2025-06-01	616031
43	2025-07-01	644493
44	2025-08-01	640803
45	2025-09-01	611277
46	2025-10-01	642315
47	2025-11-01	640201
48	2025-12-01	716924

Table 1 shows the monthly retail sales values used in the analysis. The data cover four full years, giving enough observations to study changes over time and seasonal patterns.

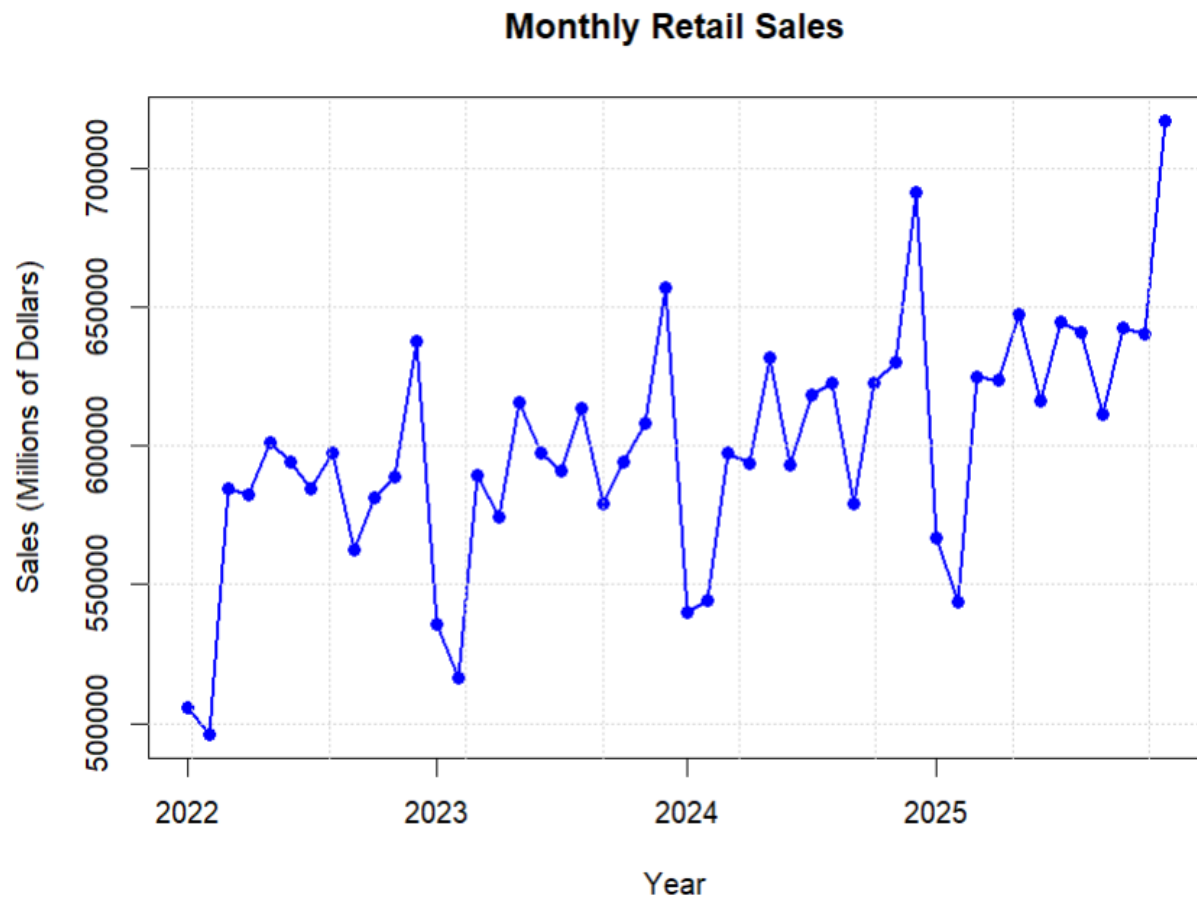


Figure 1: Monthly Retail Sales from 2022 to 2025

Figure 1 shows how monthly retail sales changed during the study period. Overall, sales increase over time, showing an upward trend. The figure also shows similar changes in the same months each year. For example, sales usually increase toward the end of the year and decrease at the beginning of the year. This may show that there is a seasonal pattern in the data, which will be studied later in the project.

Forecasting Methods

1. Naive Method

The Naive Method is one of the simplest forecasting methods. It uses the actual value from the previous period as the forecast for the next period. In simple terms, the most recent value is used to predict the next value.

$$F_t = A_{t-1}$$

where F_t is the forecast for period t , and A_{t-1} is the actual value from the previous period.

Table 2: Naive Method Forecast Accuracy

Method	MAD	MSE	MAPE
Naive	36303.78	2317620273	6.061%

The Naive Method gave a MAD of 36,303.78, an MSE of 2,317,620,272.78, and a MAPE of 6.06%. These results will be used as a reference to compare the performance of other forecasting methods.

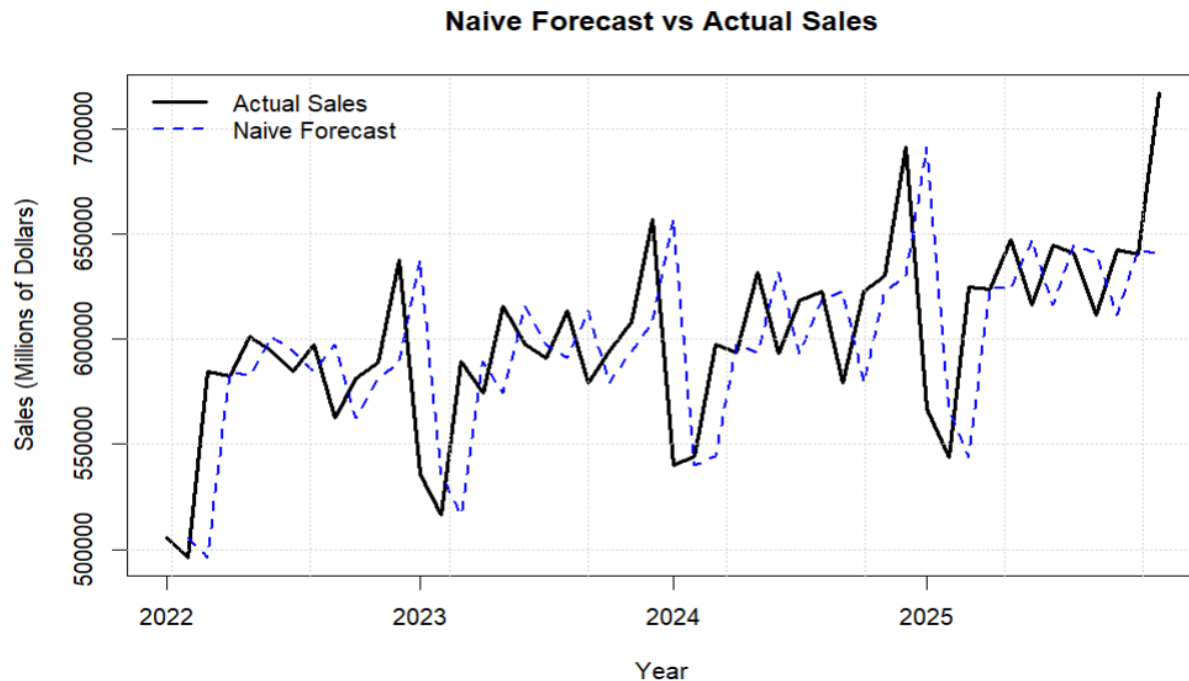


Figure 2: Actual Sales and Naive Forecast

Figure 2 compares the actual monthly sales with the Naive forecast. The Naive Method uses the actual sales from the previous month to predict the next month. Therefore, the forecast generally follows the changes in the actual sales, but with a one-month delay.

The Naive Method was used as a simple reference method to show the basic forecasting performance. It was not included in the final comparison of the forecasting methods.

2. Moving Average

The Moving Average method uses the average of a fixed number of previous values to forecast the next period. In this project, three values of (n) were tested: 3 months, 6 months, and 12 months. This was done to see how the number of months used in the average affects the accuracy of the forecast.

$$MA_n = \frac{A_{t-1} + A_{t-2} + \dots + A_{t-n}}{n}$$

where n is the number of previous periods used in the average.

Table 3: Moving Average Results

Method	n	MAD	MSE	MAPE
Moving Average	3	34525.03	1920181912	5.779%
Moving Average	6	31103.29	1552143601	5.174%
Moving Average	12	28566.06	1346533628	4.693%

The results show that the 12-month Moving Average had the lowest MAD value of 28,566.06. It also had the lowest MSE and MAPE compared with the other two values. Therefore, (n=12) gave the most accurate results among the three Moving Average settings.

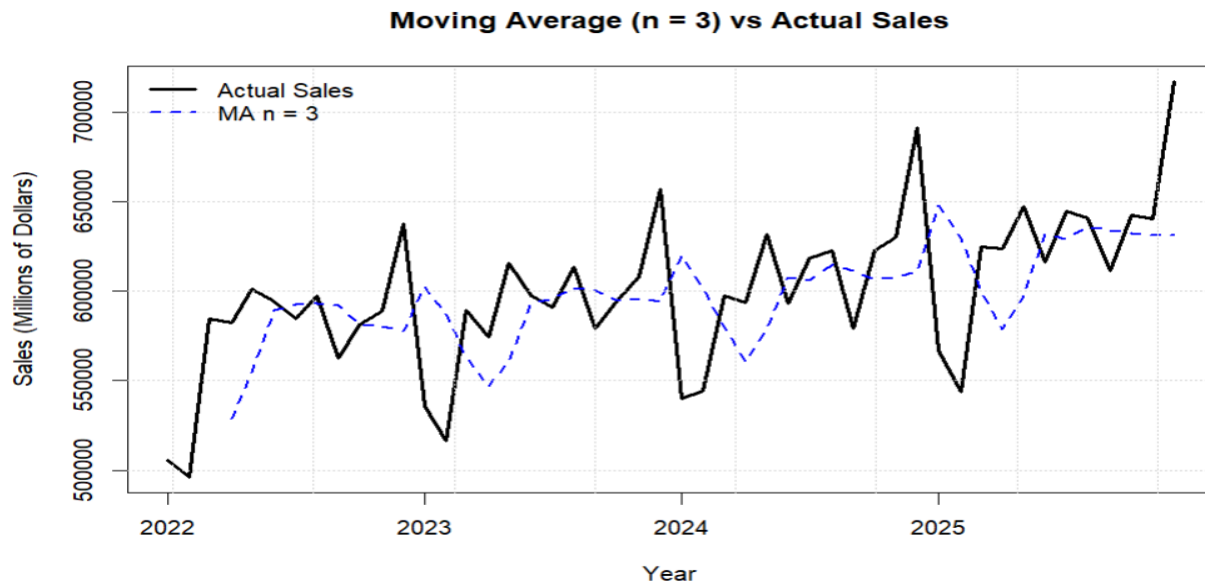


Figure 3: Actual Sales vs 3-Month Moving Average

The 3-month Moving Average reacts quickly to changes in sales because it uses the latest three months. However, there are still some clear differences between the forecast and the actual monthly sales.

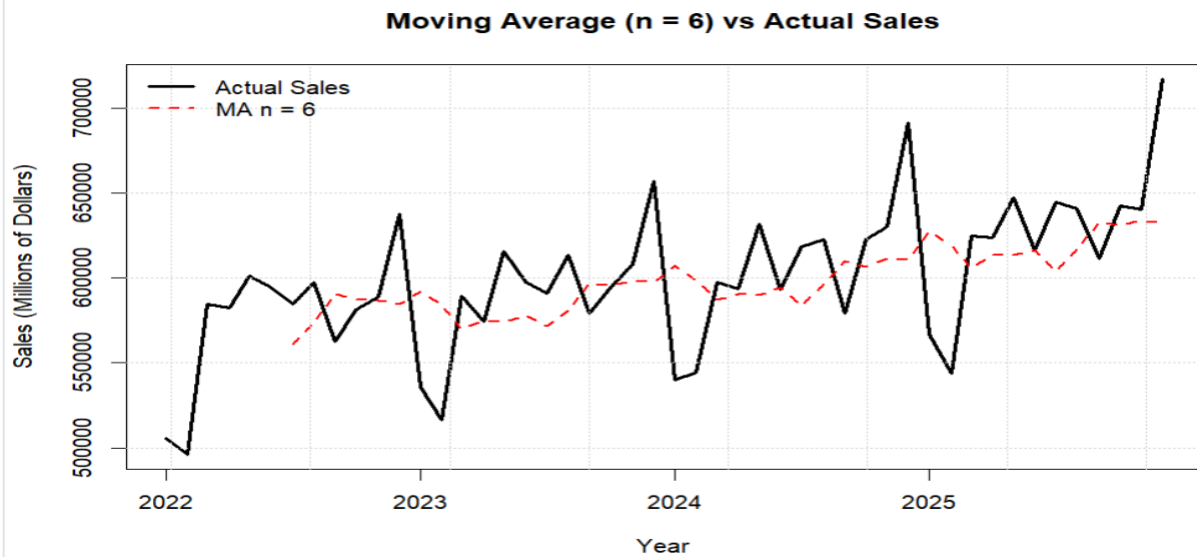


Figure 4: Actual Sales vs 6-Month Moving Average

The 6-month Moving Average gives a smoother forecast than the 3-month Moving Average. It follows the general changes in sales but responds more slowly to short-term changes.

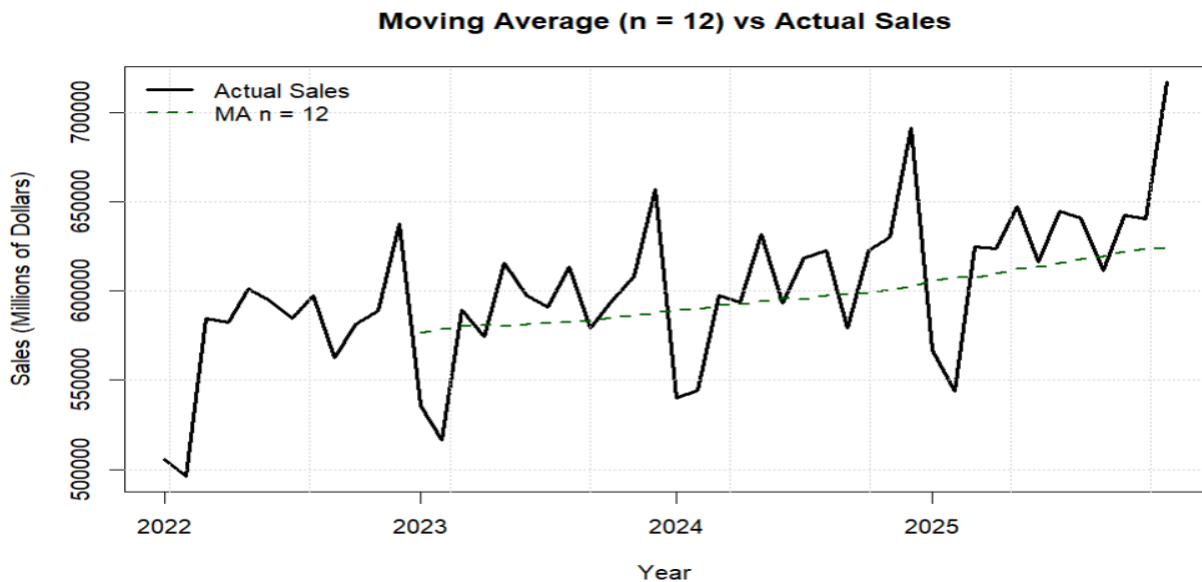


Figure 5: Actual Sales vs 12-Month Moving Average

The 12-month Moving Average gives the smoothest forecast among the three settings. It shows the overall pattern of the data and has the lowest MAD, MSE, and MAPE values.

Based on the accuracy measures and the graph, the 12-month Moving Average performed best among the three tested values.

3. Weighted Moving Average

The Weighted Moving Average method is similar to the Moving Average method, but it gives different weights to past values. Recent values usually get higher weights because they are more important for predicting the next period. In this project, three different sets of weights were tested to compare their forecasting accuracy.

$$WMA = w_1A_{t-1} + w_2A_{t-2} + w_3A_{t-3}$$

where w_1 , w_2 and w_3 are the weights assigned to the previous observations, and the total of the weights equals 1.

The following weight combinations were tested:

(0.50, 0.30, 0.20), (0.60, 0.30, 0.10), and (0.70, 0.20, 0.10).

Table 4: Weighted Moving Average Results

Method	Weights	MAD	MSE	MAPE
Weighted Moving Average	0.50, 0.30, 0.20	34983.15	1882678999	5.859%
Weighted Moving Average	0.60, 0.30, 0.10	35662.23	1946890592	5.977%
Weighted Moving Average	0.70, 0.20, 0.10	35085.14	1954987450	5.869%

The results show that the weight combination (0.50, 0.30, 0.20) had the lowest MAD value of 34,983.15. It also had the lowest MSE of 1,882,678,999 and the lowest MAPE of 5.859%. Therefore, the weight combination (0.50, 0.30, 0.20) gave the most accurate forecast among the three Weighted Moving Average settings.

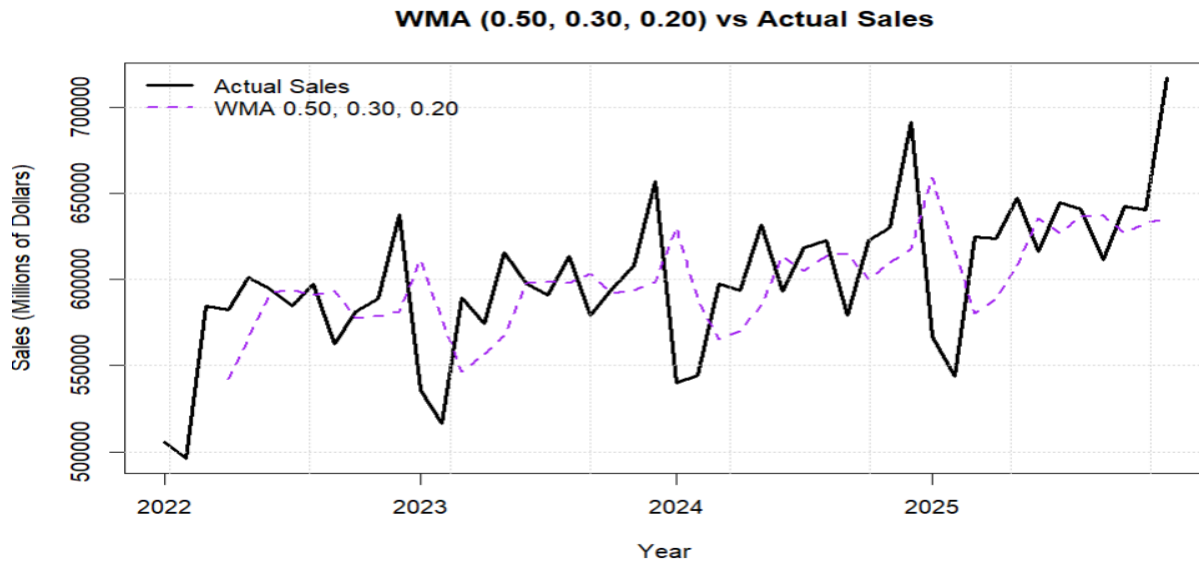


Figure 6: Actual Sales vs Weighted Moving Average (0.50, 0.30, 0.20)

Figure 6 shows the forecast using weights of 0.50, 0.30, and 0.20. This combination gave the lowest MAD, MSE, and MAPE values among the three tested combinations.

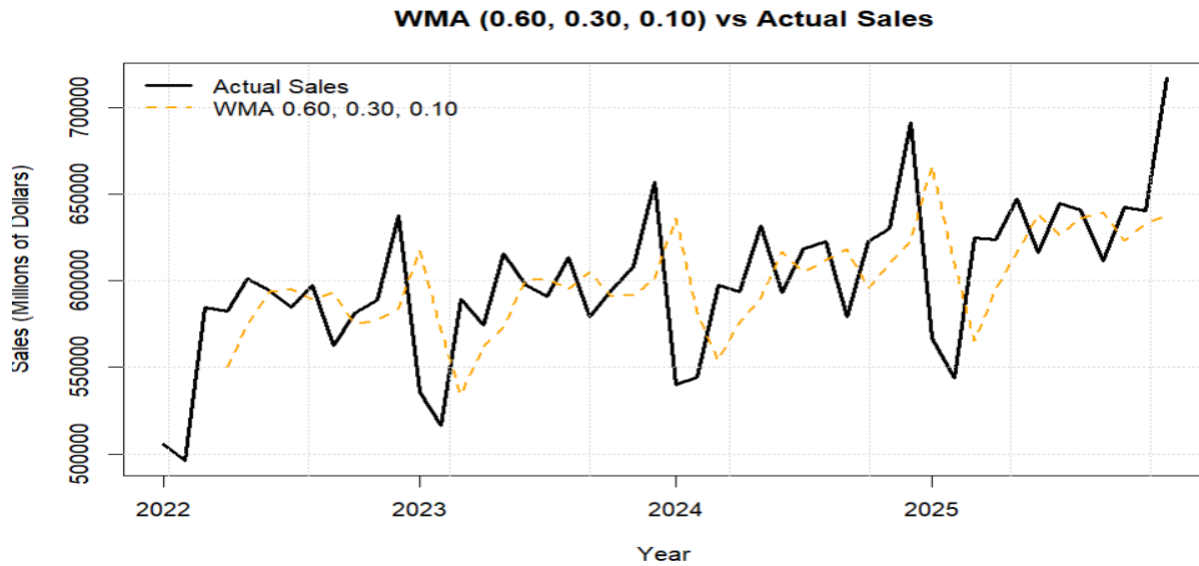


Figure 7: Actual Sales vs Weighted Moving Average (0.60, 0.30, 0.10)

Figure 7 shows the forecast using weights of 0.60, 0.30, and 0.10. Giving more weight to the most recent value did not improve the forecast accuracy compared with the first case.

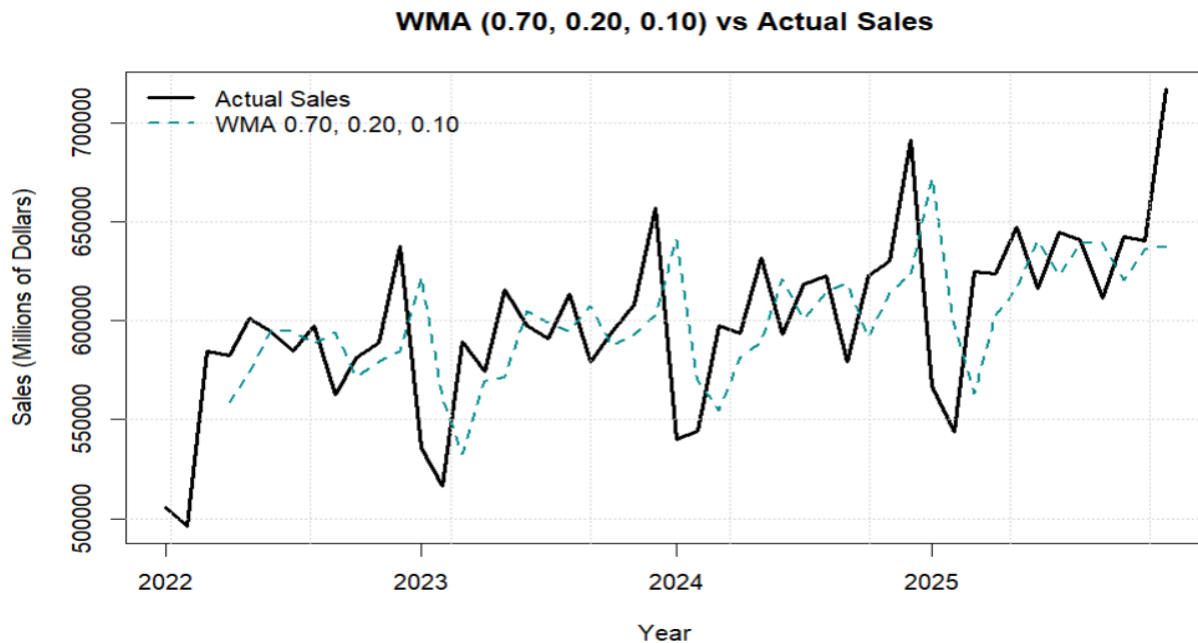


Figure 8: Actual Sales vs Weighted Moving Average (0.70, 0.20, 0.10)

Figure 8 shows the forecast using weights of 0.70, 0.20, and 0.10. Although this case gives the highest weight to the most recent value, its error measures were higher than those of the first case.

Overall, the results show that giving more weight to the most recent value did not improve the forecasting accuracy for this dataset.

4. Simple Exponential Smoothing

The Simple Exponential Smoothing method gives more importance to recent values when forecasting the next period. It uses a smoothing constant called alpha (α), which controls the weight given to the most recent actual value. In this project, three different alpha values were tested: 0.10, 0.30, and 0.50.

$$F_t = \alpha A_{t-1} + (1 - \alpha)F_{t-1}$$

where α is the smoothing constant, A_{t-1} is the previous actual value, and F_{t-1} is the previous forecast.

Table 5: Simple Exponential Smoothing Results

Method	Alpha	MAD	MSE	MAPE
Exponential Smoothing	0.1	34068.47	1625184232	5.555%
Exponential Smoothing	0.3	31778.15	1569162121	5.293%
Exponential Smoothing	0.5	33634.80	1750687402	5.626%

The results show that alpha = 0.30 had the lowest MAD value of 31,778.15. It also had the lowest MSE of 1,569,162,121 and the lowest MAPE of 5.293%. Therefore, alpha = 0.30 gave the most accurate forecast among the three Exponential Smoothing settings.

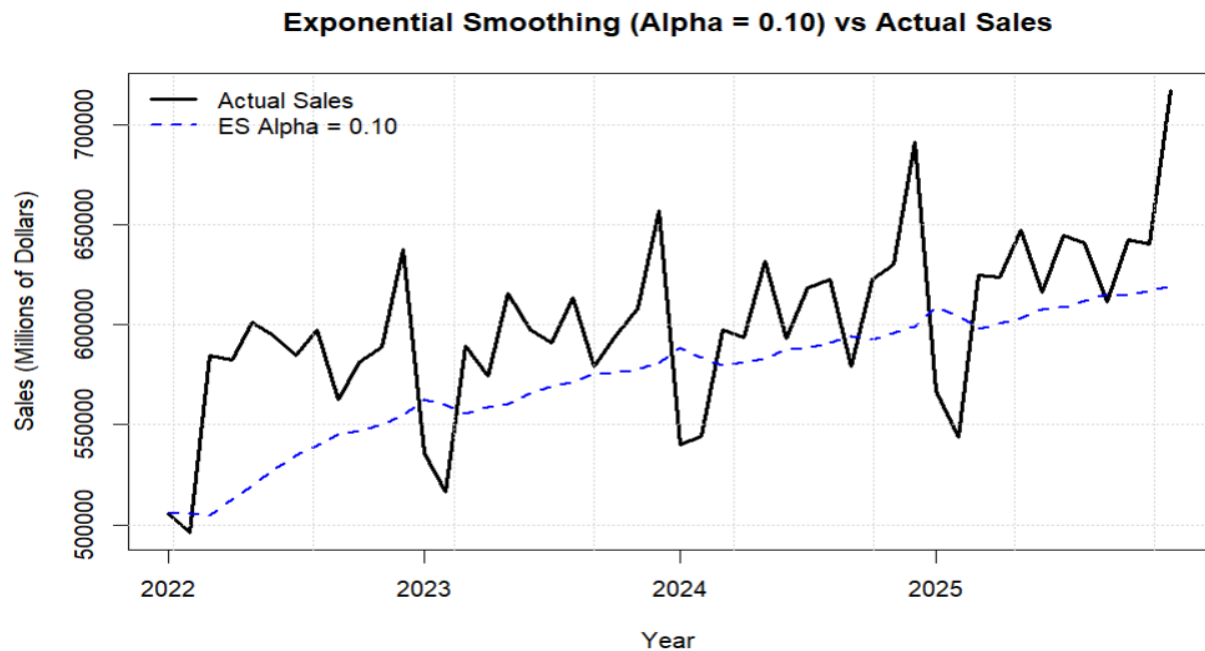


Figure 9: Actual Sales vs Exponential Smoothing ($\alpha = 0.10$)

Figure 9 shows the forecast using alpha = 0.10. This low alpha value gives a smoother forecast and reacts more slowly to changes in the actual sales values.

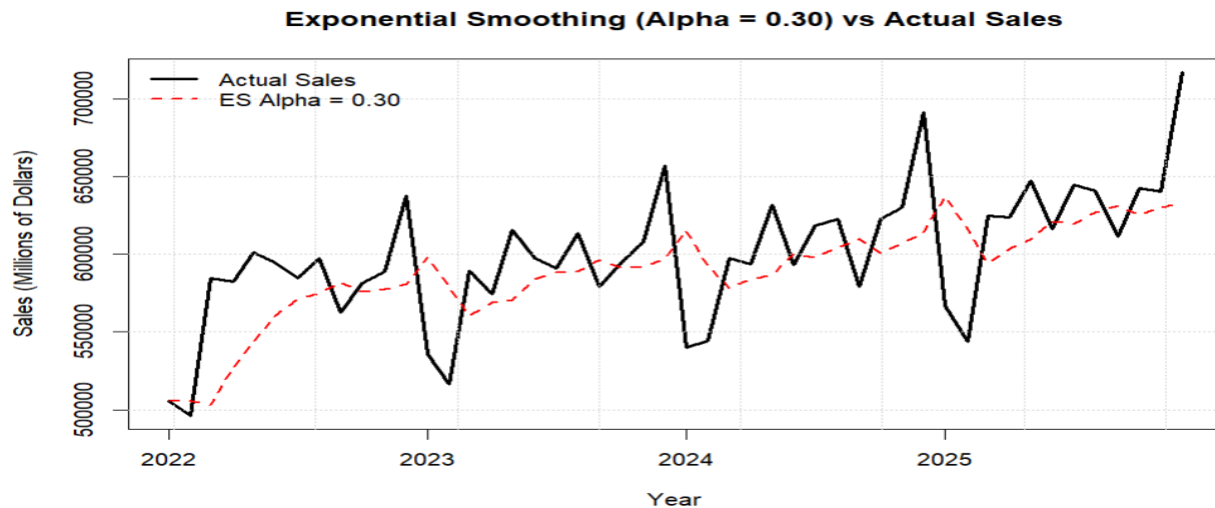


Figure 10: Actual Sales vs Exponential Smoothing ($\alpha = 0.30$)

Figure 10 shows the forecast using $\alpha = 0.30$. This setting gives a good balance between smoothing the data and responding to changes in sales. It also produced the lowest forecast errors among the three tested values.

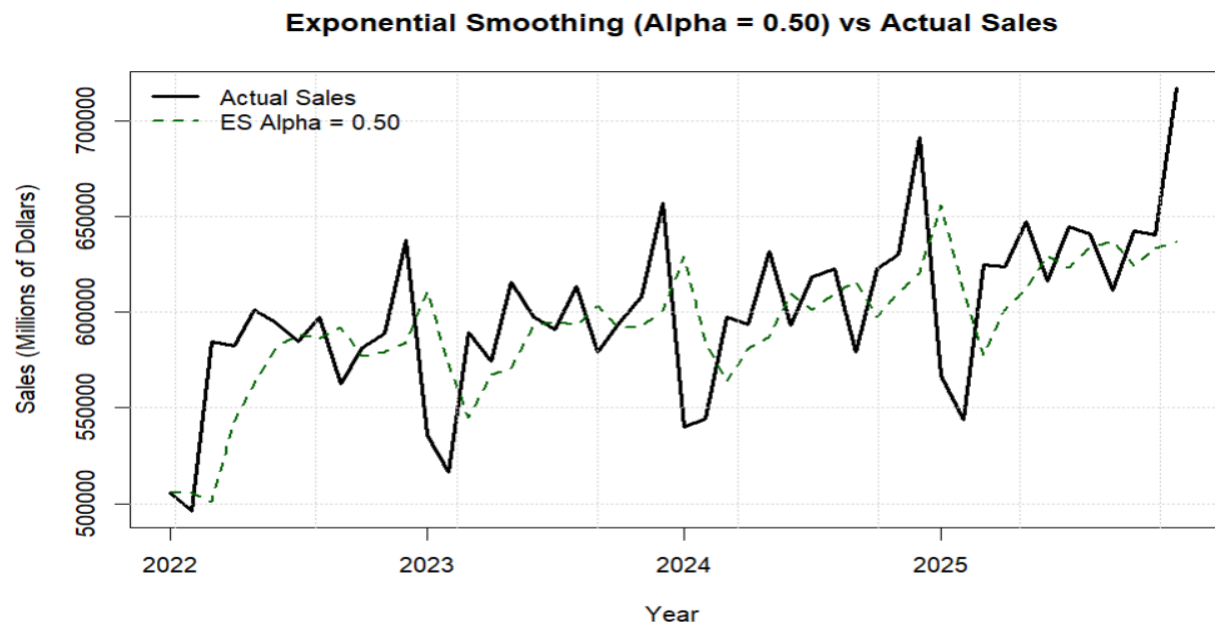


Figure 11: Actual Sales vs Exponential Smoothing ($\alpha = 0.50$)

Figure 11 shows the forecast using $\alpha = 0.50$. This higher alpha value makes the forecast respond more quickly to recent changes, but the error measures were higher than those for $\alpha = 0.30$.

Overall, increasing alpha did not always improve the forecasting accuracy. The best result was obtained with $\alpha = 0.30$.

5. Trend-Adjusted Exponential Smoothing (FIT)

Trend-Adjusted Exponential Smoothing is used when the data have a trend over time. This method is an extension of Simple Exponential Smoothing because it also includes a trend component. It uses two smoothing constants: alpha (α) for the level of the data and beta (β) for the trend. In this project, four different combinations of alpha and beta were tested to compare their forecasting accuracy.

$$F_t = \alpha A_{t-1} + (1 - \alpha)(F_{t-1} + T_{t-1})$$

$$T_t = \beta(F_t - F_{t-1}) + (1 - \beta)T_{t-1}$$

$$FIT_t = F_t + T_t$$

where F_t represents the smoothed forecast, T_t represents the trend, α is the smoothing constant for the level, and β is the smoothing constant for the trend.

Four combinations were tested: ($\alpha = 0.10$, $\beta = 0.10$), ($\alpha = 0.30$, $\beta = 0.20$), ($\alpha = 0.50$, $\beta = 0.30$), and ($\alpha = 0.70$, $\beta = 0.50$).

Table 6: Trend-Adjusted Exponential Smoothing Results

Method	Alpha	Beta	MAD	MSE	MAPE
FIT	0.1	0.1	27005.41	1413062158	4.574%
FIT	0.3	0.2	31923.15	1772419746	5.383%
FIT	0.5	0.3	36486.31	2116401922	6.141%
FIT	0.7	0.5	41260.78	2866273366	6.926%

The results show that Case 1, with $\alpha = 0.10$ and $\beta = 0.10$, had the lowest MAD value of 27,005.41. It also had the lowest MSE of 1,413,062,158 and the lowest MAPE of 4.574%. Therefore, Case 1 gave the most accurate forecast among the four Trend-Adjusted Exponential Smoothing settings.

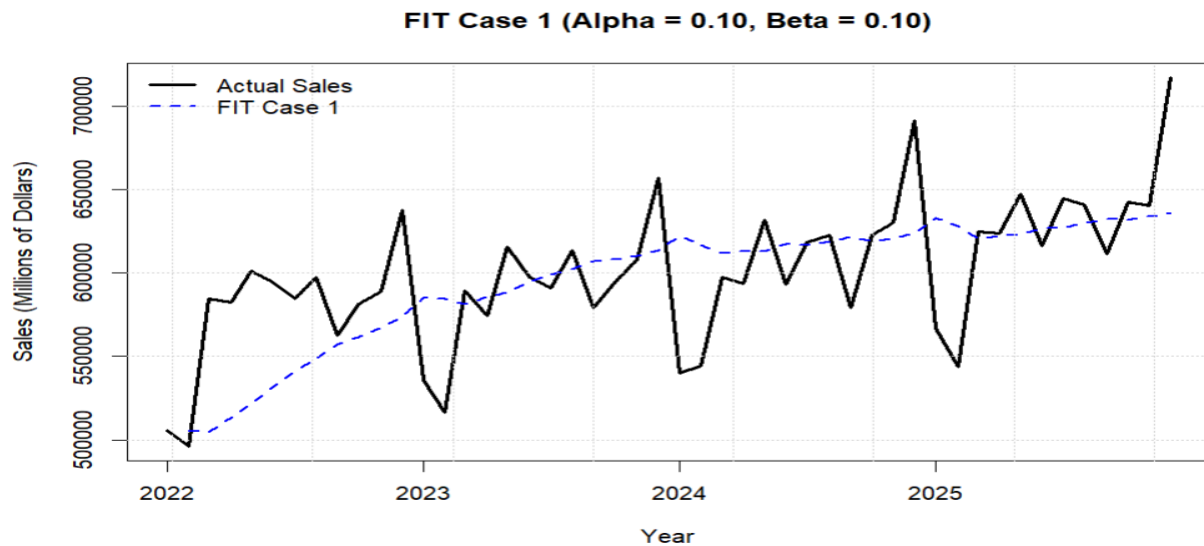


Figure 12: Actual Sales vs FIT ($\alpha = 0.10$, $\beta = 0.10$)

Figure 12 shows the forecast using $\alpha = 0.10$ and $\beta = 0.10$. This case had the lowest forecasting errors among the four tested combinations.

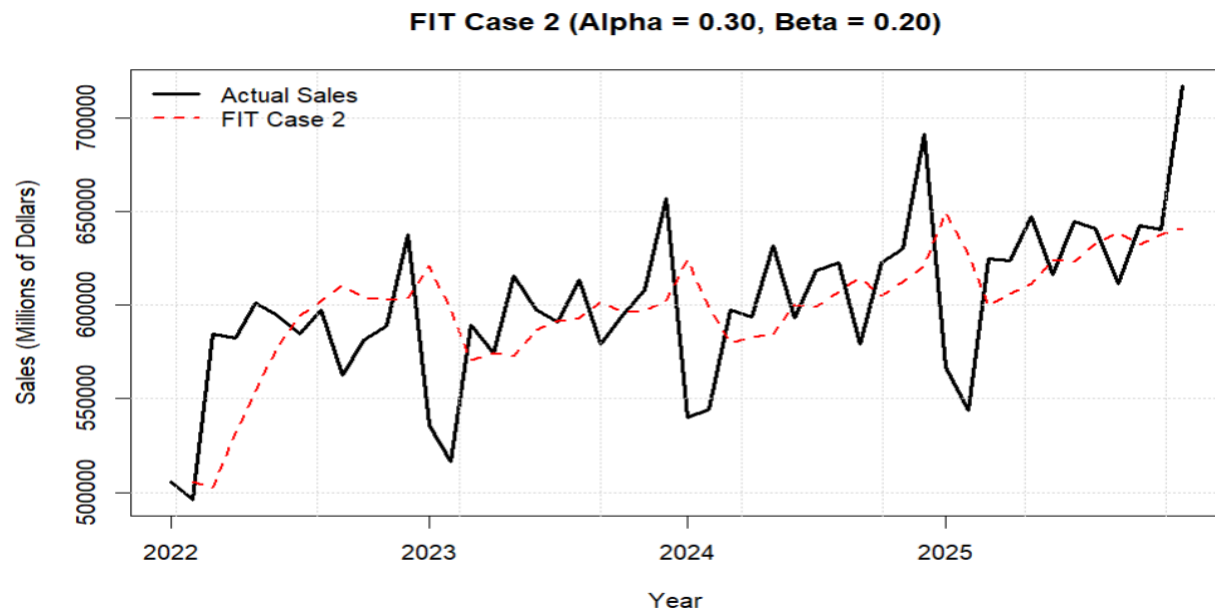


Figure 13: Actual Sales vs FIT ($\alpha = 0.30, \beta = 0.20$)

Figure 13 shows the forecast using $\alpha = 0.30$ and $\beta = 0.20$. The forecast responds more strongly to changes in the data, but its error measures were higher than those of Case 1.

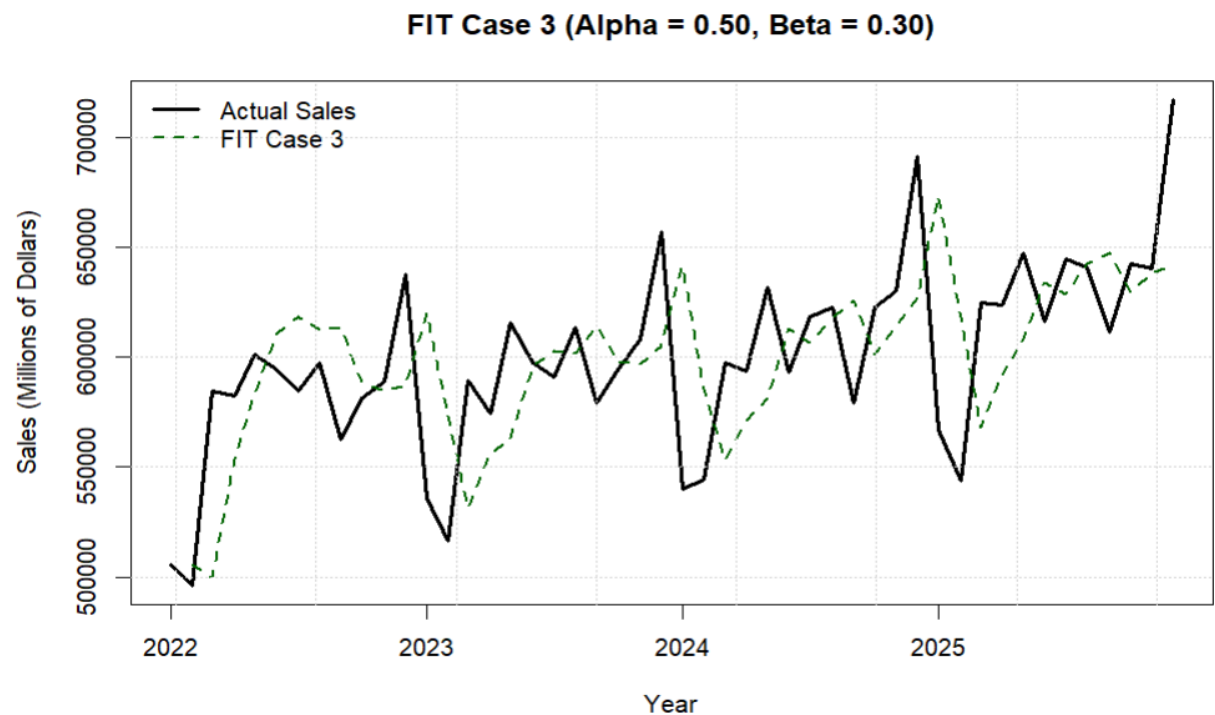


Figure 14: Actual Sales vs FIT ($\alpha = 0.50, \beta = 0.30$)

Figure 14 shows the forecast using $\alpha = 0.50$ and $\beta = 0.30$. Increasing both smoothing constants resulted in higher forecasting errors compared with the first two cases.

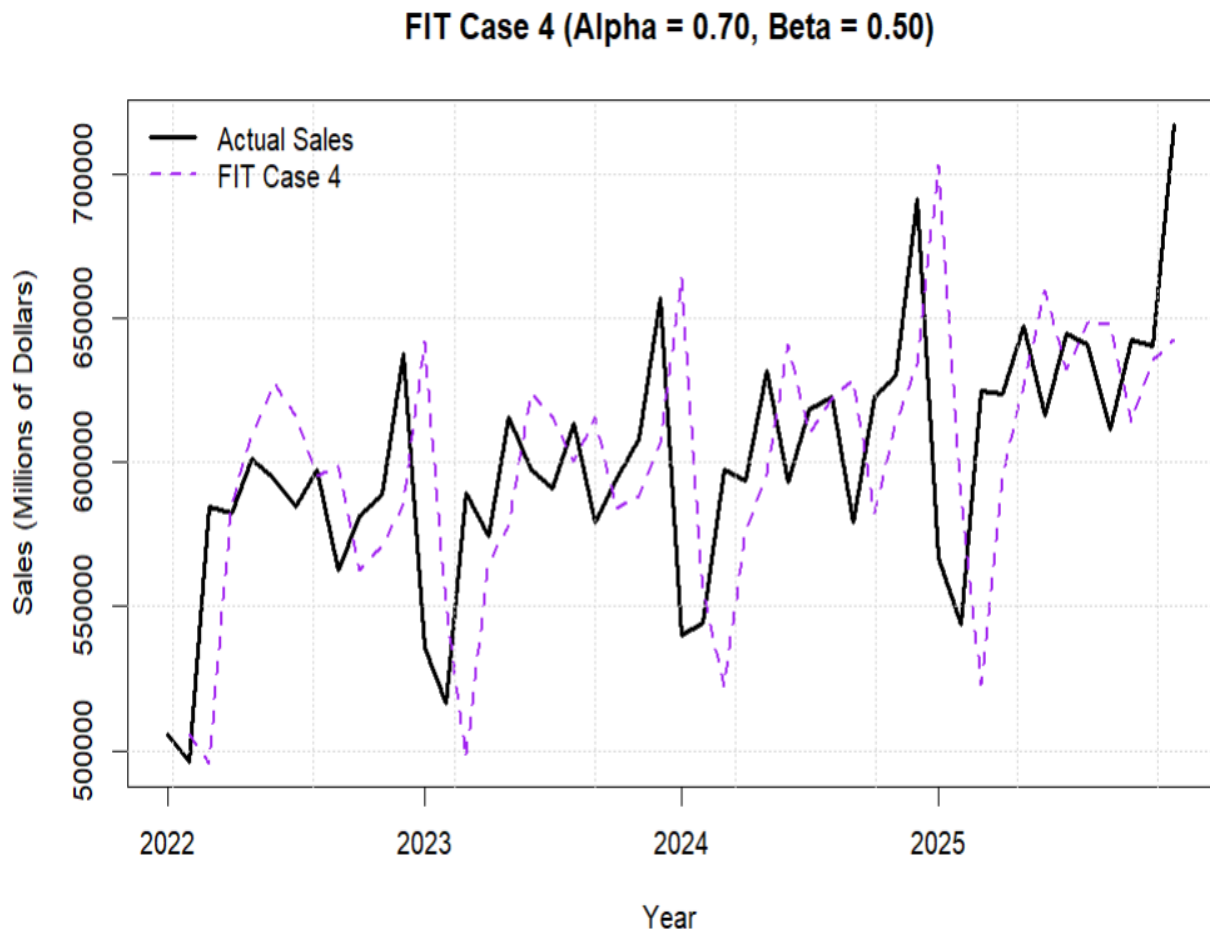


Figure 15: Actual Sales vs FIT ($\alpha = 0.70$, $\beta = 0.50$)

Figure 15 shows the forecast using $\alpha = 0.70$ and $\beta = 0.50$. This combination had the highest MAD, MSE, and MAPE values among the four tested cases.

Overall, increasing the values of α and β did not improve the forecasting accuracy for this dataset. The lowest values of α and β , both equal to 0.10, gave the best FIT results.

6. Trend Projection

Trend Projection is used when the data show an overall upward or downward movement over time. This method fits a straight trend line to the historical data and uses time as the independent variable to estimate sales. In this project, a linear trend model was used to study the overall direction of monthly retail sales.

$$Y_t = a + bt$$

where a is the intercept, b is the slope, and t represents the time period.

$$\text{Sales} = 554933.44 + 1816.36 * t$$

The positive slope of 1,816.36 means that monthly retail sales increased on average during the study period.

Table 7: Trend Projection Results

Method	Intercept	Slope	MAD	MSE	MAPE
Trend Projection	554933.4	1816.36	24611.49	1180715391	4.139%

The Trend Projection method produced a MAD of 24,611.49, an MSE of 1,180,715,391, and a MAPE of 4.139%. The results show that the linear trend represents the general upward movement of the sales data.

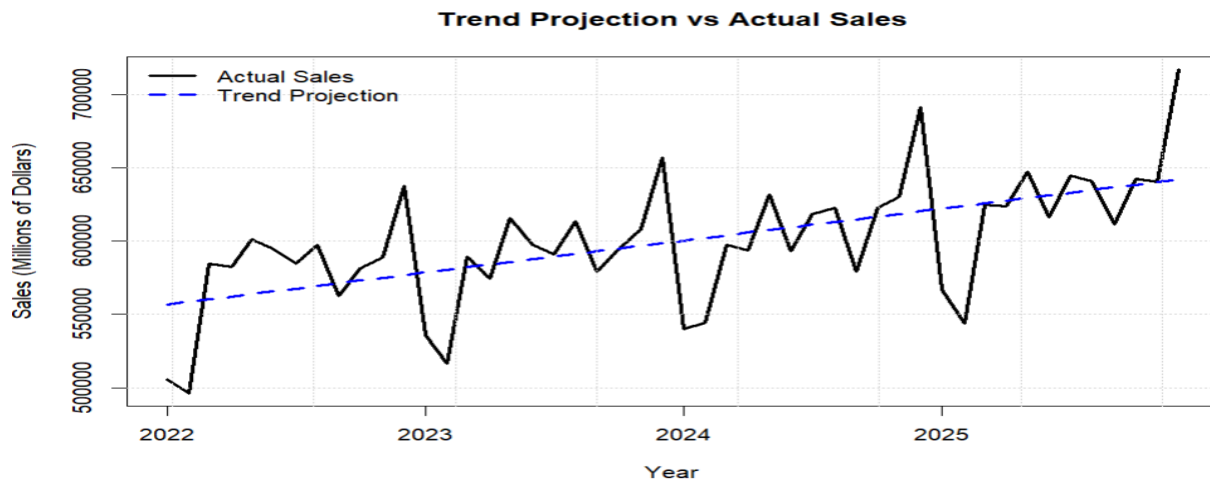


Figure 16: Actual Sales vs Trend Projection

Figure 16 compares the actual monthly sales with the estimated trend line. The trend line increases gradually over time, while the actual sales move above and below the line because of monthly changes and seasonal effects.

Trend Projection was used to identify the general trend in the data and to support the seasonality analysis. It was not included in the final comparison of forecasting methods.

Seasonality Analysis

Seasonality refers to repeated patterns that happen during the same periods over time. Since the data are monthly, the seasonal pattern was analyzed across the 12 months of the year. First, the trend effect was removed from the actual sales values, and then seasonal indices were calculated for each month.

A seasonal index greater than 1 means that sales are higher than the normal trend level for that month, while an index below 1 means that sales are lower than the normal trend level.

Table 8: Monthly Seasonal Indices

Month	Seasonal Index	Seasonal Effect
January	0.911	-8.89%
February	0.888	-11.15%
March	1.011	1.1%
April	0.998	-0.17%
May	1.046	4.64%
June	1.004	0.41%
July	1.016	1.57%
August	1.028	2.78%
September	0.966	-3.42%
October	1.007	0.72%
November	1.016	1.57%
December	1.109	10.86%

The seasonal indices show clear differences between the months. Months with an index above 1 have sales above the general trend, while months with an index below 1 have sales below the trend. The results show that sales are generally lower at the beginning of the year and higher toward the end of the year.

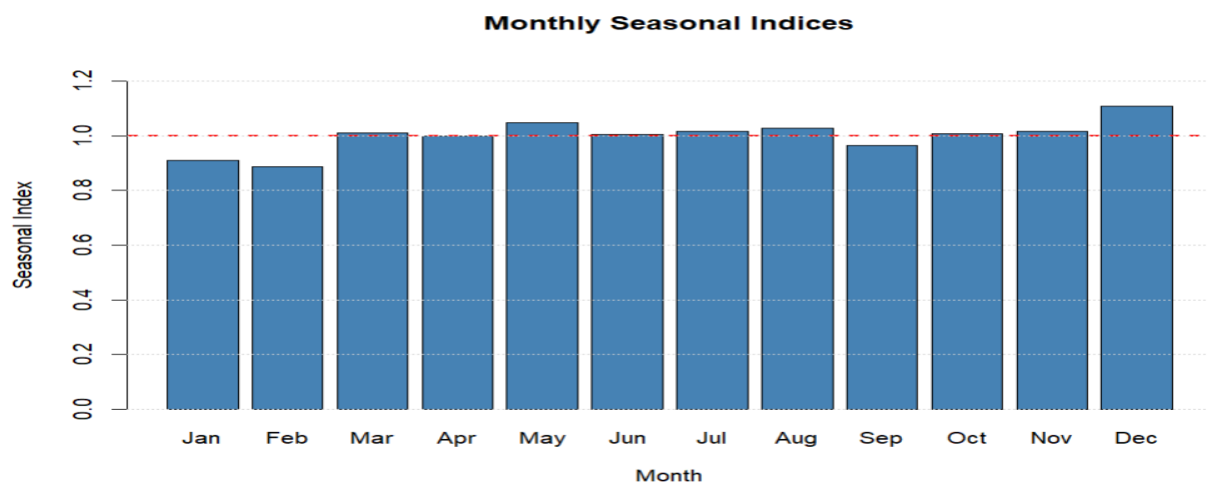


Figure 17: Monthly Seasonal Indices

Figure 17 shows the seasonal pattern across the 12 months. December has the highest seasonal index, while January and February have the lowest values. This shows that the monthly sales data have a clear seasonal pattern.

Table 9: Seasonal Ratios and Monthly Seasonal Indices

Month	Ratio_2022	Ratio_2023	Ratio_2024	Ratio_2025	Seasonal Index
January	0.908	0.926	0.899	0.911	0.911
February	0.888	0.89	0.904	0.872	0.888
March	1.043	1.013	0.99	0.999	1.011
April	1.036	0.983	0.98	0.994	0.998
May	1.066	1.051	1.04	1.028	1.046
June	1.051	1.017	0.973	0.976	1.004
July	1.03	1.003	1.012	1.018	1.016
August	1.049	1.038	1.015	1.009	1.028
September	0.985	0.977	0.942	0.96	0.966
October	1.014	0.999	1.01	1.006	1.007
November	1.024	1.02	1.019	1	1.016
December	1.105	1.098	1.114	1.116	1.109

Table 9 shows the seasonal ratios for each month across the four years. These ratios compare the actual sales values with the estimated trend values. Ratios above 1 mean that sales were higher than the trend level, while ratios below 1 mean that sales were lower than the trend level. The average pattern across the four years was then used to calculate the monthly seasonal index.

The results show a consistent seasonal pattern across the years. January and February have ratios below 1 in all four years, while December has ratios above 1 in every year. This supports the presence of seasonality in the monthly sales data.

Since the data show both trend and seasonality, the seasonal indices were combined with the trend values to develop the Combined Trend and Seasonal Forecast.

Combined Trend and Seasonal Forecast

The Combined Trend and Seasonal Forecast combines the estimated trend with the monthly seasonal index. This allows the forecast to show both the general movement of sales over time and the seasonal pattern of each month.

$$\text{Combined Forecast} = \text{Trend Forecast} \times \text{Seasonal Index}$$

The trend value shows the general direction of sales over time, while the seasonal index adjusts the forecast based on the expected seasonal effect for each month.

Table 10: Combined Trend and Seasonal Forecast – 2025

Month	Sales	Trend Forecast	Seasonal Index	Seasonal Trend Forecast
Jan-25	566662	622138.9	0.911	566800.6
Feb-25	543970	623955.27	0.888	554379.7
Mar-25	624954	625771.6	1.011	632656.5
Apr-25	623756	627588.0	0.998	626499.2
May-25	647306	629404.4	1.046	658613.1
Jun-25	616031	631220.7	1.004	633783.7
Jul-25	644493	633037.1	1.016	642950.3
Aug-25	640803	634853.5	1.028	652482.1
Sep-25	611277	636669.8	0.966	614900.6
Oct-25	642315	638486.2	1.007	643067.4
Nov-25	640201	640302.5	1.016	650382.0
Dec-25	716924	642118.9	1.109	711833.1

Table 10 shows how the trend forecast was adjusted using the monthly seasonal index for 2025. Months with seasonal indices above 1 had higher combined forecasts, while months with indices below 1 had lower combined forecasts. For example, December has a high seasonal index of 1.109, which increased the trend forecast from 642,118.9 to 711,833.1.

Table 11: Combined Trend and Seasonal Forecast Accuracy

Method	MAD	MSE	MAPE
Combined Trend and Seasonal Forecast	6694.12	67018280	1.114%

The Combined Trend and Seasonal Forecast produced a MAD of 6,694.12, an MSE of 67,018,280, and a MAPE of 1.114%. These results show that the combined model followed the actual sales values closely during the study period.

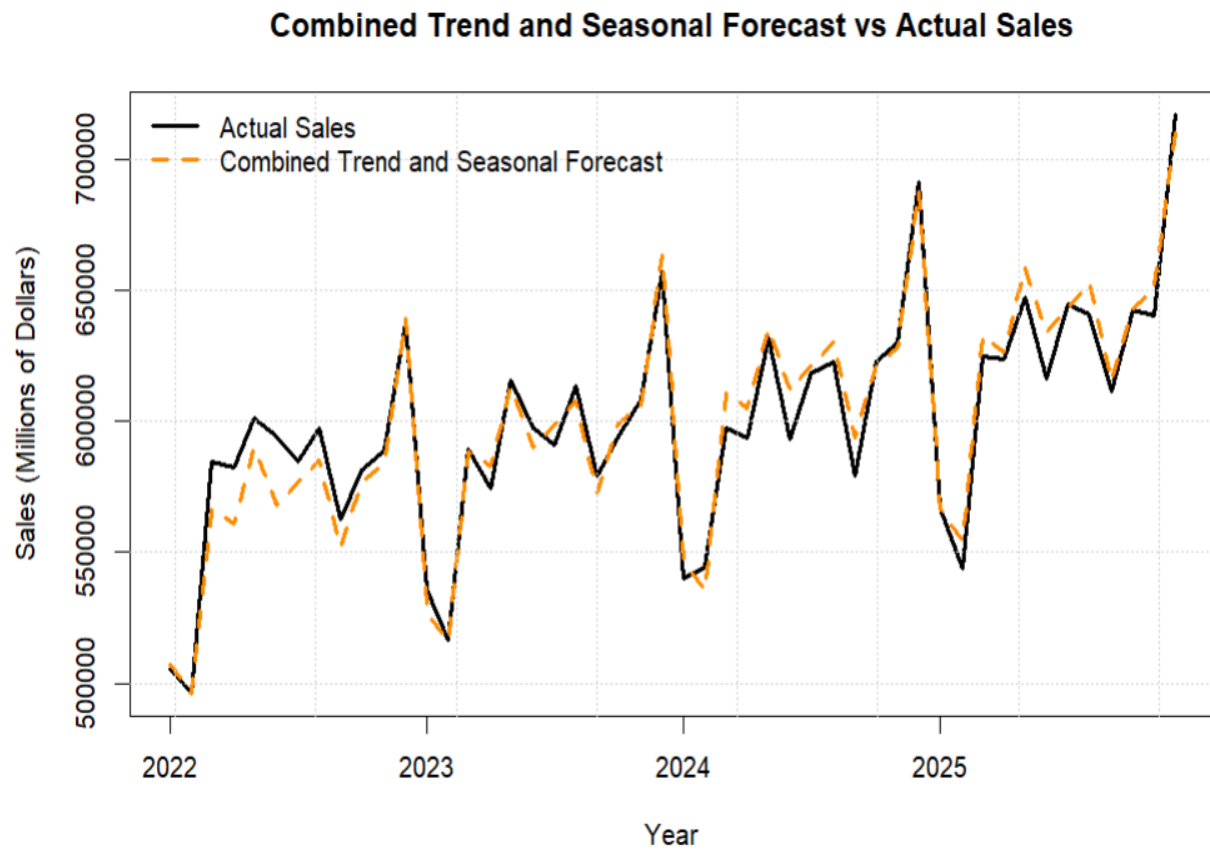


Figure 18: Combined Trend and Seasonal Forecast vs Actual Sales

Figure 18 shows that the Combined Trend and Seasonal Forecast follows the actual sales values closely throughout the study period. The forecast captures both the upward trend and the repeated seasonal changes in monthly sales. The small differences between the two lines are consistent with the low MAD, MSE, and MAPE values.

Although this method produced low forecasting errors, it was used as part of the trend and seasonality analysis and was not included in the final comparison of the main forecasting methods.

Final Comparison of Forecasting Methods

The final comparison was conducted among the four main forecasting methods: Moving Average, Weighted Moving Average, Simple Exponential Smoothing, and Trend-Adjusted Exponential Smoothing (FIT). All tested parameter settings were compared using MAD, MSE, and MAPE. The main criterion for selecting the most accurate forecasting method was the lowest MAD value.

Table 12: Final Comparison of All Tested Forecasting Methods

Method	Setting	MAD	MSE	MAPE
Moving Average	n = 3	34525.03	1920181912	5.779%
Moving Average	n = 6	31103.29	1552143601	5.174%
Moving Average	n = 12	28566.06	1346533628	4.693%
Weighted Moving Average	0.50, 0.30, 0.20	34983.15	1882678999	5.859%
Weighted Moving Average	0.60, 0.30, 0.10	35662.23	1946890592	5.977%
Weighted Moving Average	0.70, 0.20, 0.10	35085.14	1954987450	5.869%
Exponential Smoothing	Alpha = 0.10	34068.47	1625184232	5.555%
Exponential Smoothing	Alpha = 0.30	31778.15	1569162121	5.293%
Exponential Smoothing	Alpha = 0.50	33634.80	1750687402	5.626%
FIT	Alpha = 0.10, Beta = 0.10	27005.41	1413062158	4.574%
FIT	Alpha = 0.30, Beta = 0.20	31923.15	1772419746	5.383%
FIT	Alpha = 0.50, Beta = 0.30	36486.31	2116401922	6.141%
FIT	Alpha = 0.70, Beta = 0.50	41260.78	2866273366	6.926%

Table 12 compares all 13 tested forecasting settings. The results show that the forecasting accuracy changes depending on the method and the parameter values used. The lowest MAD among all tested cases was obtained by the Trend-Adjusted Exponential Smoothing method with $\alpha = 0.10$ and $\beta = 0.10$.

Table 13: Best Setting from Each Forecasting Method

Method	Setting	MAD	MSE	MAPE
Moving Average	n = 12	28566.06	1346533628	4.693%
Weighted Moving Average	0.50, 0.30, 0.20	34983.15	1882678999	5.859%
Exponential Smoothing	Alpha = 0.30	31778.15	1569162121	5.293%
FIT	Alpha = 0.10, Beta = 0.10	27005.41	1413062158	4.574%

Table 13 summarizes the best setting for each forecasting method. Based on MAD, the FIT method with $\alpha = 0.10$ and $\beta = 0.10$ had the lowest error value of 27,005.41. It also had the lowest MAPE of 4.574%. The 12-month Moving Average had the lowest MSE of 1,346,533,628. However, since MAD was used as the main evaluation criterion, FIT with $\alpha = 0.10$ and $\beta = 0.10$ was selected as the most accurate forecasting method.

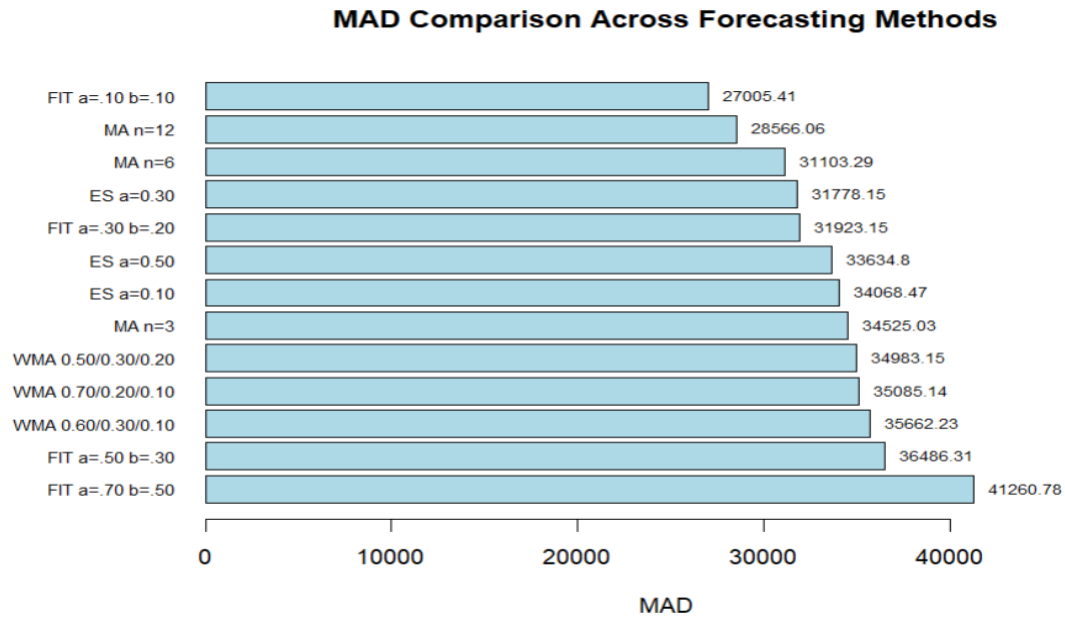


Figure 19: MAD Comparison Across Forecasting Methods

Figure 19 compares the MAD values for all tested forecasting settings. Lower MAD values mean better forecasting accuracy. The Trend-Adjusted Exponential Smoothing method with $\alpha = 0.10$ and $\beta = 0.10$ had the lowest MAD value of 27,005.41, followed by the 12-month Moving Average with a MAD of 28,566.06. The highest MAD was obtained by FIT with $\alpha = 0.70$ and $\beta = 0.50$. Based on MAD, FIT with $\alpha = 0.10$ and $\beta = 0.10$ gave the most accurate forecast among the tested cases.

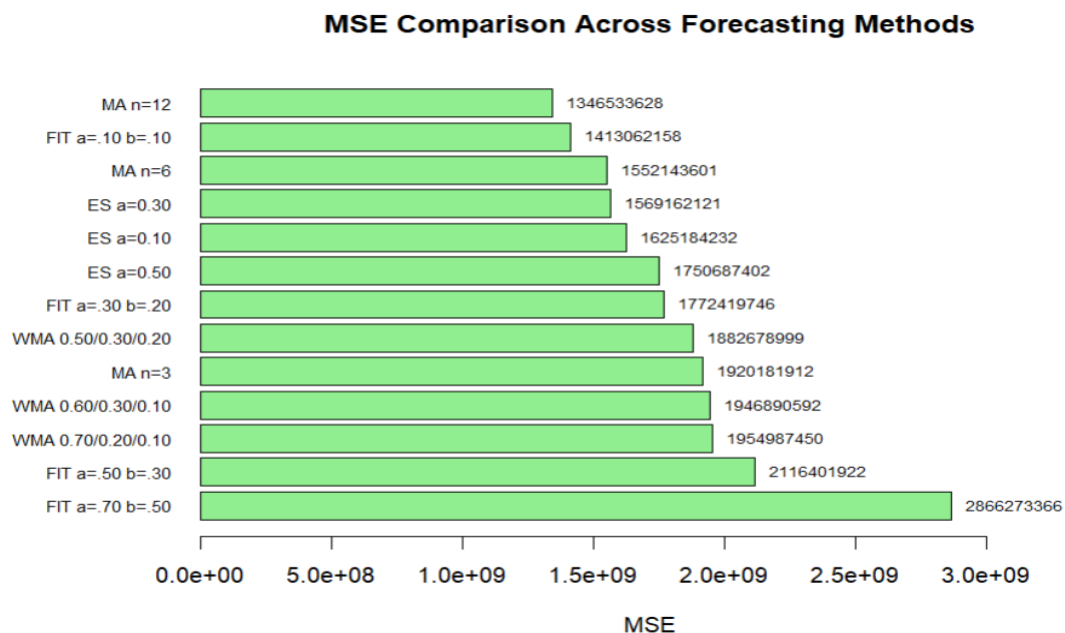


Figure 20: MSE Comparison Across Forecasting Methods

Figure 20 compares the MSE values for all tested forecasting settings. Lower MSE values mean better forecasting performance. The 12-month Moving Average had the lowest MSE value of 1,346,533,628, followed by FIT with $\alpha = 0.10$ and $\beta = 0.10$. The highest MSE was obtained by FIT with $\alpha = 0.70$ and $\beta = 0.50$. Although the 12-month Moving Average had the lowest MSE, MAD was used as the main criterion for selecting the best forecasting method.

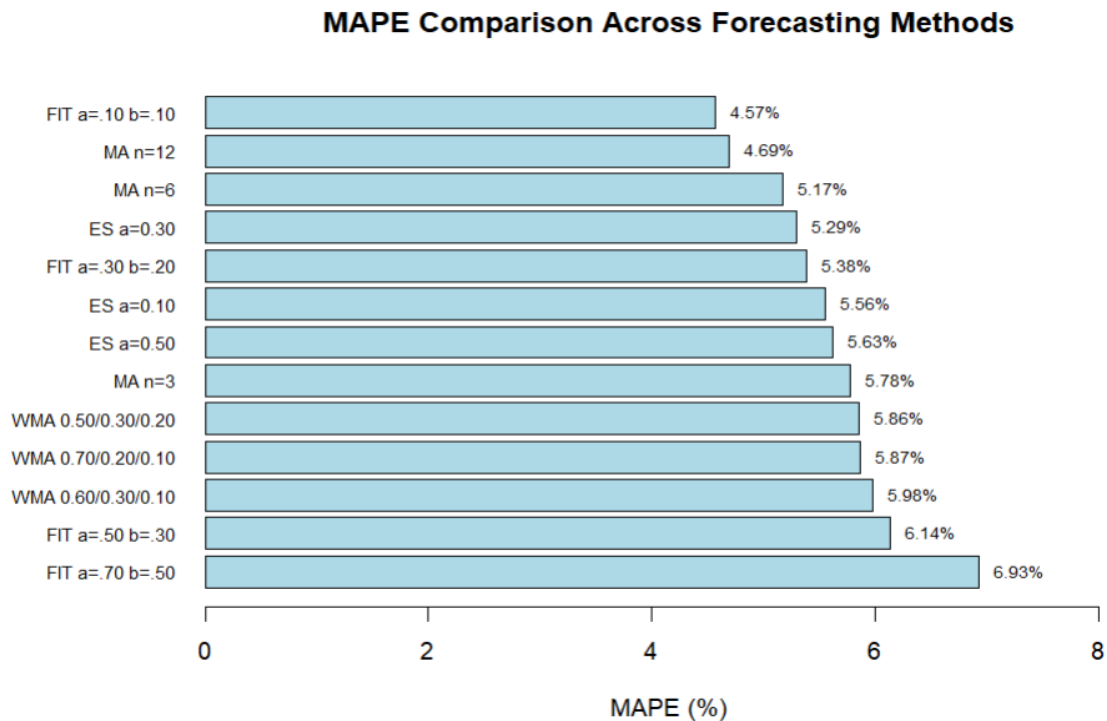


Figure 21: MAPE Comparison Across Forecasting Methods

Figure 21 compares the MAPE values for all tested forecasting settings. Lower MAPE values mean better forecasting accuracy. FIT with $\alpha = 0.10$ and $\beta = 0.10$ had the lowest MAPE value of 4.57%, followed by the 12-month Moving Average with 4.69%. The highest MAPE was obtained by FIT with $\alpha = 0.70$ and $\beta = 0.50$. These results support the good performance of FIT with $\alpha = 0.10$ and $\beta = 0.10$.

Overall, the final comparison shows that FIT with $\alpha = 0.10$ and $\beta = 0.10$ had the lowest MAD and MAPE among the tested forecasting methods. The 12-month Moving Average had the lowest MSE, but MAD was used as the main criterion for selecting the best method. Therefore, FIT with $\alpha = 0.10$ and $\beta = 0.10$ was selected as the preferred forecasting method for this dataset.

Conclusion and Recommendation

This project used different time series forecasting methods on monthly retail sales data from January 2022 to December 2025. The analysis was done using R to organize the data, apply the forecasting methods, calculate the accuracy measures, and create the charts. The methods used were the Naive Method, Moving Average, Weighted Moving Average, Simple Exponential Smoothing, Trend-Adjusted Exponential Smoothing (FIT), and Trend Projection. The final comparison focused on Moving Average, Weighted Moving Average, Simple Exponential Smoothing, and FIT. Different parameter values were tested, and the results were compared using MAD, MSE, and MAPE.

The results showed that FIT with $\alpha = 0.10$ and $\beta = 0.10$ had the lowest MAD value of 27,005.41 and the lowest MAPE value of 4.574% among the methods in the final comparison. The 12-month Moving Average had the lowest MSE value of 1,346,533,628. Since MAD was used as the main criterion, FIT with $\alpha = 0.10$ and $\beta = 0.10$ was selected as the preferred method.

The seasonality analysis also showed a clear pattern in the monthly sales. Sales were usually lower at the beginning of the year, especially in January and February, and higher toward the end of the year. December had the highest seasonal index. The Combined Trend and Seasonal Forecast also showed that including seasonality helped the forecast follow the actual monthly sales more closely.

Based on the final comparison, FIT with $\alpha = 0.10$ and $\beta = 0.10$ is recommended as the preferred forecasting method among the methods compared. However, seasonal patterns should also be considered when forecasting monthly sales because they have an effect on the data.

References

U.S. Census Bureau. (2026). *Retail Sales: Retail Trade (MRTSSM44000USN)*. St. Louis: Federal Reserve Bank of St. Louis. Retrieved from <https://fred.stlouisfed.org/series/MRTSSM44000USN>